Chapter-3

1. a) $F(x, y, z) = \Sigma(2, 3, 6, 7)$

$x \backslash yz$	00	01	11	10
0			1	1
1			1	1

$F = y$

b) $F(A, B, C, D) = \Sigma(7, 13, 14, 15)$

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	4	5	1 7	6
11	12	1 13	1 15	1 14
10	8	9	11	10

$F = BCD + AB$

c) $F(A, B, C, D) = \Sigma(4, 6, 7, 15)$

$AB \backslash CD$	00	01	11	10
00				
01	1		1	1
11			1	
10				

$F = BCD + A'BD'$

d) $F(w, x, y, z) = \Sigma(2, 3, 12, 13, 14, 15)$

$wx \backslash yz$	00	01	11	10
00			1	1
01				
11	1	1	1	1
10				

$F = w'x'y + wx$

2. a) $xy + x'y'z' + x'yz'$

$x \backslash yz$	00	01	11	10
0	1			1
1			1	1

$F = xy + x'z'$

b) $x'.\ A'B + BC' + B'C'$

$A \backslash BC$	00	01	11	10
0	1		1	1
1	1			1

$F = A'B + C'$

c) $a'b' + bc + a'bc'$

$a \backslash bc$	00	01	11	10
0	1	1	1	1
1			1	

$F = x' + yz$

d) $xy'z + xyz' + x'yz + xyz$

$x \backslash yz$	00	01	11	10
0			1	
1		1	1	1

$= yz + x$

3. a) $D(A'+B) + B'(C+AD)$

$= DA' + DB + B'C + B'AD$

AB \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	
11	1	1	1	
10	1	1	1	1

$= D + B'C$

b) $ABD + A'C'D' + A'B + A'CD' + AB'D'$

AB \ CD	00	01	11	10
00	1			1
01	1	1	1	1
11		1	1	
10	1			1

$= B'D' + A'B + B'D'$

c) $k'lm' + k'm'n + klm'n' + lmn'$

kl \ mn	00	01	11	10
00		1		
01	1	1		
11	1			1
10				1

=

d) $A'B'C'D' + AC'D' + B'CD' + A'BCD + BC'D$

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11	1	1		
10	1			1

$= A'BD + B'D' + ABC'$

e) $x'z + w'xy' + w(x'y + xy')$

wx \ yz	00	01	11	10
00	1			1
01	1			
11		1		
10		1	1	

$= w'z' + wyz + wx'z$

4. a) $F(A,B,C,D,E) = \Sigma(0,1,4,5,16,17,21,25,29)$

For $A=0$ ie A' 0-15

BC \ DE	00	01	11	10
00	1	1		
01	1	1		
11				
10				

For $A=1$ ie A 16-31

BC \ DE	00	01	11	10
00	1	1		
01		1		
11		1		
10		1		

$F = (A,B,C,D,E) = A'B'D' + AD'E + B'C'D'$

b. $BDE + B'C'D + CDE + A'B'CE + A'B'C + B'C'D'E$

For $A=0$ ie, A'

BC \ DE	00	01	11	10
00		1		1
01	1	1	1	
11				
10			1	

For $A=1$ ie A

BC \ DE	00	01	11	10
00		1		1
01			1	
11				
10			1	

AB \ CDE	000	001	011	010	110	111	101	100
00	1		1	1	1	1	1	1
01			1			1		
11			1			1		
10	1		1	1		1		

$= C'DE + CDE + B'C'D + A'B'C + B'C'D'E'$

$= DE + B'C'D + A'BC' + B'C'D'E'$

C. $A'B'CE' + A'B'C'D' + B'D'E' + B'CD' + CDE' + BDE'$

For $A=0$ ie A'

BC \ DE	00	01	11	10
00	1	1		
01	1	1	1	1
11			1	1
10			1	1

For $=A$ ie A

BC \ DE	00	01	11	10
00	1			
01	1	1		
11				
10				

$F(A, B, C, DE) = B'D'E' + B'CD' + AB'D' + A'B'C + A'BD$

3.5. a) $F_1 = \pi(0,3,5,6)$

$F_2 = \pi(0,1,2,4)$

b)

$F_1 =$

x \ yz	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$F_2 =$

x \ yz	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$F_1 = x'y'z + x'yz' + xy'z' + xyz$

$F_2 = xz + xy + yz$

c)

$F_1 =$

x \ yz	00	01	11	10
0	0	1	0	1
1	1	0	1	0

x \ yz	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$F_1 = (x+y+z)(x+y'+z')(x'+y+z')(x'+y'+z)$

$F_2 = (x+y)(y+z)(x+z)$

3.6. a) $F(x,y,z) = \pi(0,1,4,5)$

x \ yz	00	01	11	10
0	0	0	1	1
1	0	0	1	1

b) $F(A,B,C,D) = \pi(0,1,2,3,4,10,11)$

x \ yz	00	01	11	10
0	0	0	0	0
	0		1	1
1	0	1	1	1

$F = (A+B)(B+C')(A+C+D)$

c) $F(w,x,y,z) = \pi(1,3,5,7,13,15)$

wx \ yz	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	1	1	1

$F = (w+z')(x'+z')$

7. a) $x'z' + y'z' + yz' + xyz$

x \ yz	00	01	11	10
0	1	0	0	1
1	1	0	1	1

i) sum of products

$= z' + xy$

ii) product of sum $= (y+z')(x+z')$

b) $(A+B'+D)(A'+B+D)(C+D)(C'+D')$

AB \ CD	00	01	11	10
00	0	1	0	1
01	0	1	0	0
11	0	1	0	1
10	0	1	0	0

i) sum of products

$= C'D + A'B'CD' + ABCD'$

ii) Product of sum

$(A'+B+D)$

$= (C+D)(C'+D')(A+B'+D)(A'+B+D')(B+C'+D')$

c) $(A'+B'+D')(A+B'+C')(A'+B+D')$

AB \ CD	00	01	11	10
00				
01				
11				
10				

i) sum of product

$= C'D + A'C' + AD' + A'B'D'$

ii) Product of sum

$= (C'+D')(A'+D')(A+B'+C')$

d) $(A'+B'+D)(A'+D')(A+B+D')(A+B'+C+D)$

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	0	0	0
10	1	0	0	1

i) sum of product

$= A'BD + A'CD' + B'C'D' + AB'D'$

ii) Product of sum

$= (A'+B')(B+D')(B'+C+D)$

e) $w'yz' + vw'z' + vw'x + v'wz + v'w'y'z'$

vw \ xyz	000	001	011	010	110	111	101	100
00	1	0	0	1	1	0	0	1
01	0	1	1	0	0	1	1	0
11	0	0	0	0	0	0	0	0
10	1	0	0	1	1	1	1	1

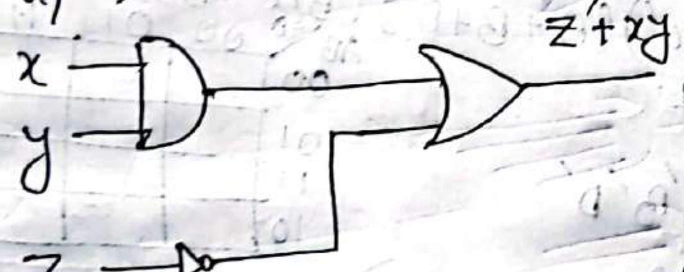
i) Sum of product

$= w'yz' + yw'z' + vw'x + v'wz + v'w'y'z'$

ii) Product of sum

$= (v'+w')(w'+y'+z')(w+x+z')(w'+y+z)(v+w+x'+z)$

8. a) i)



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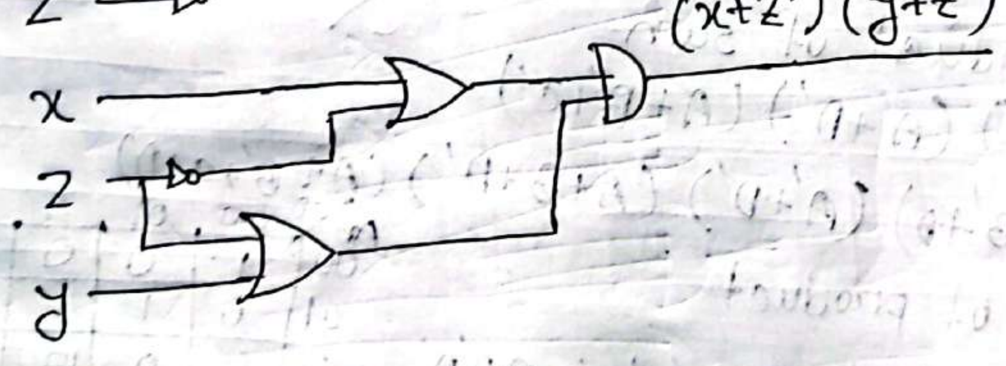
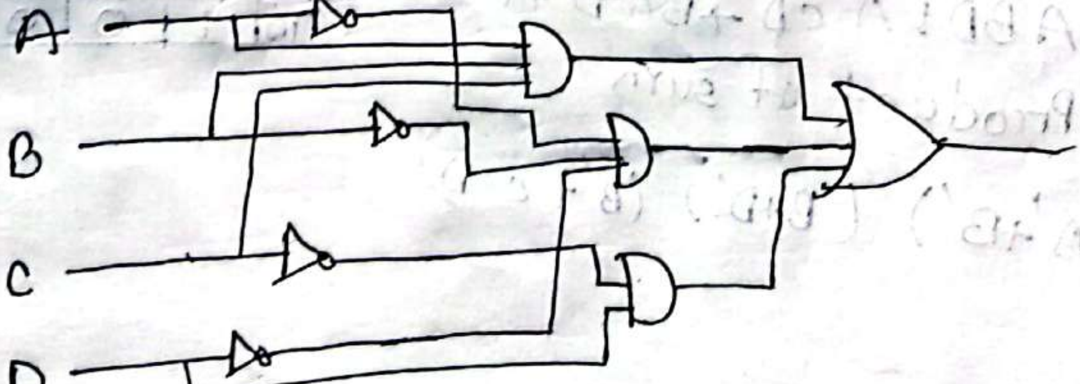


b) i)



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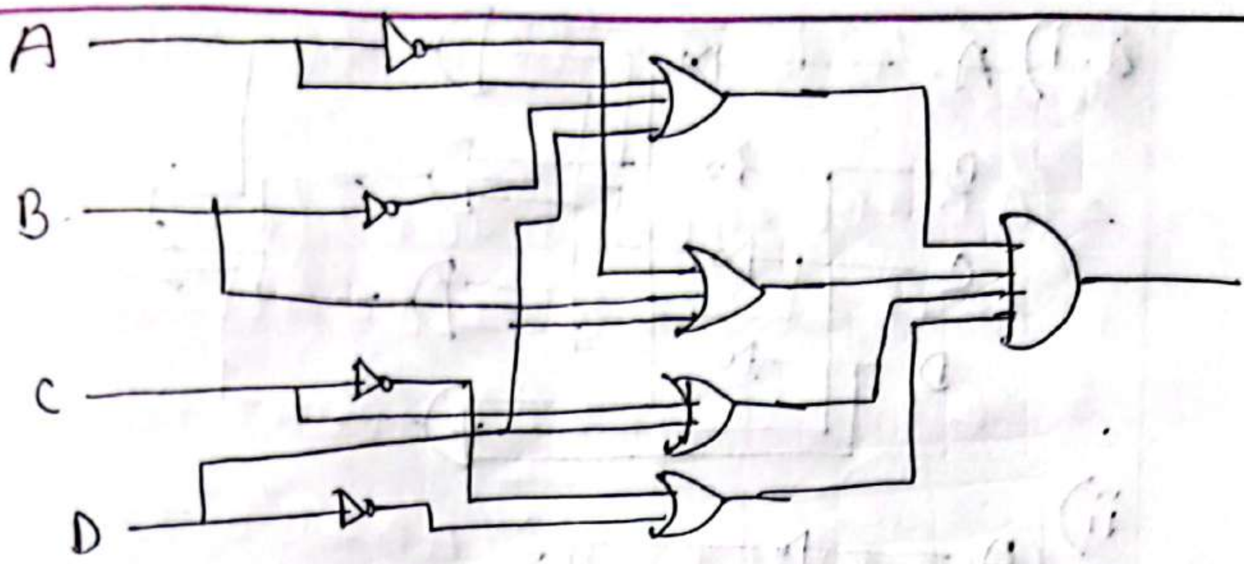
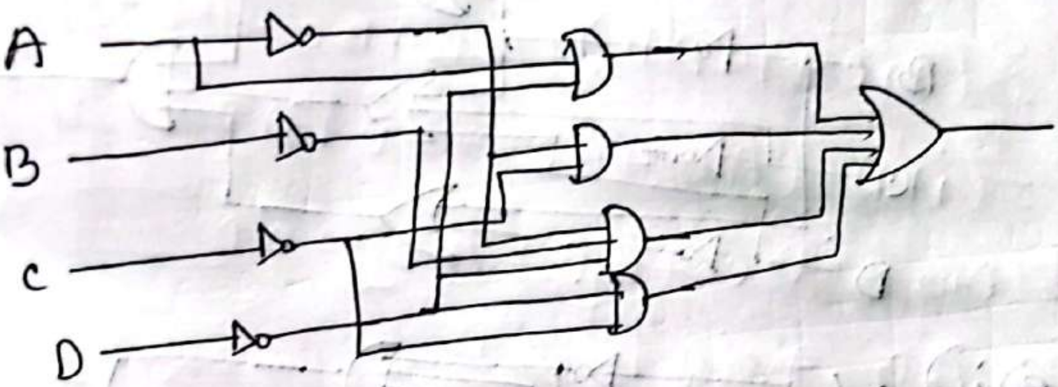
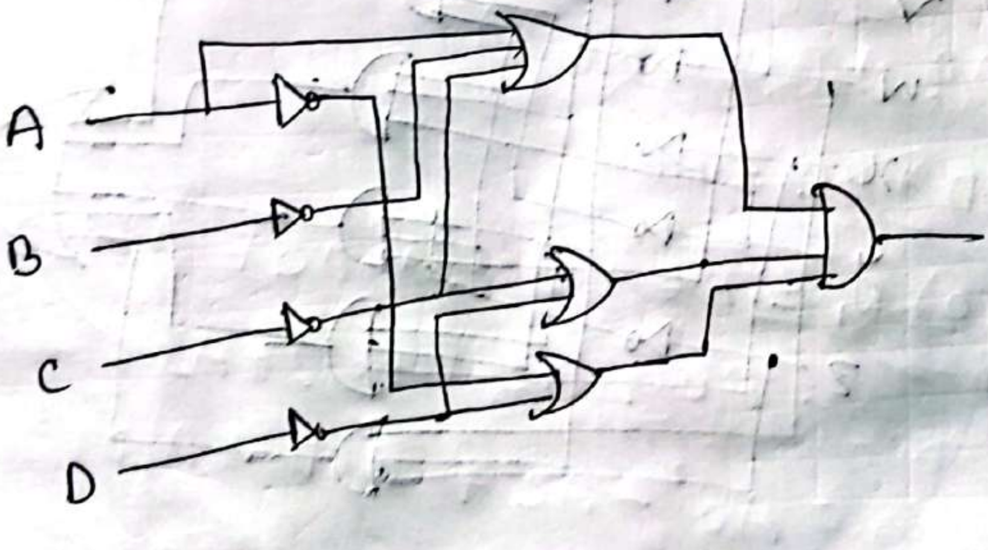


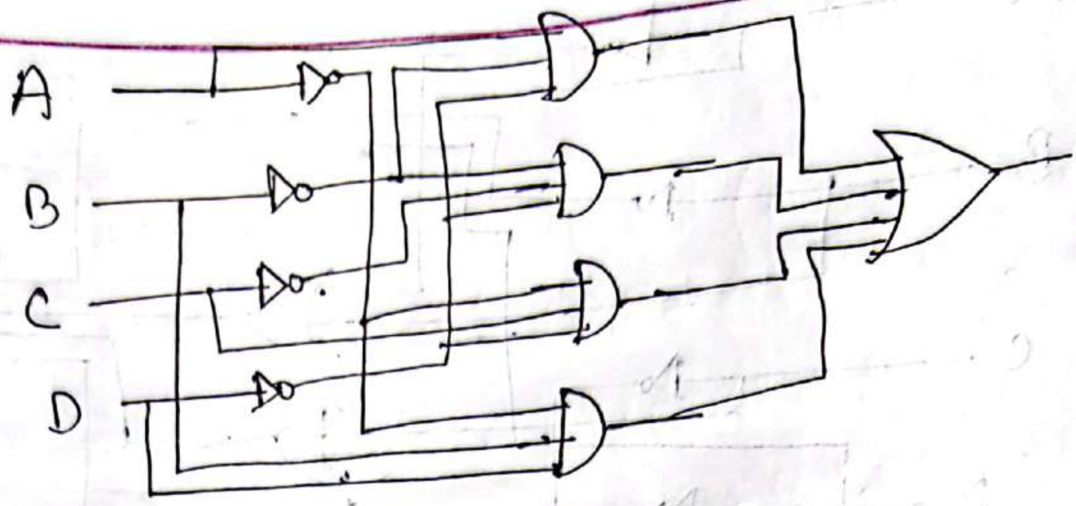
c) i.



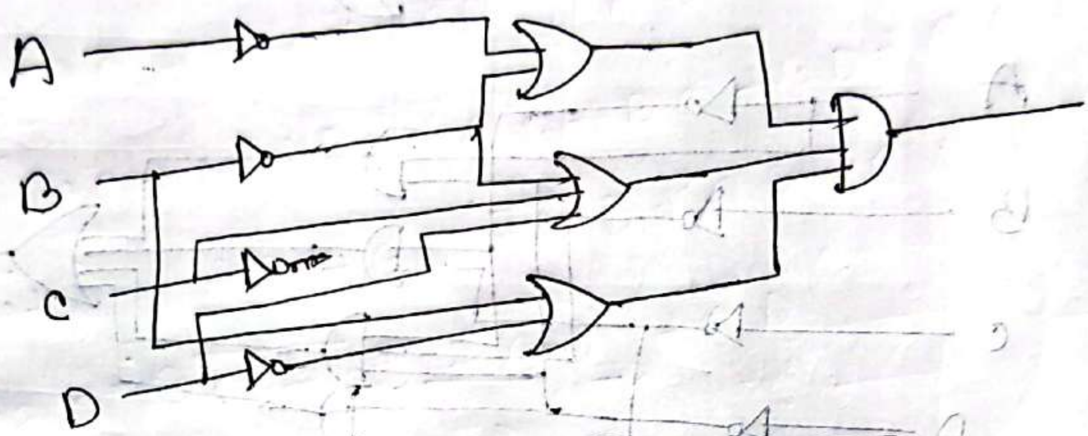
ii.



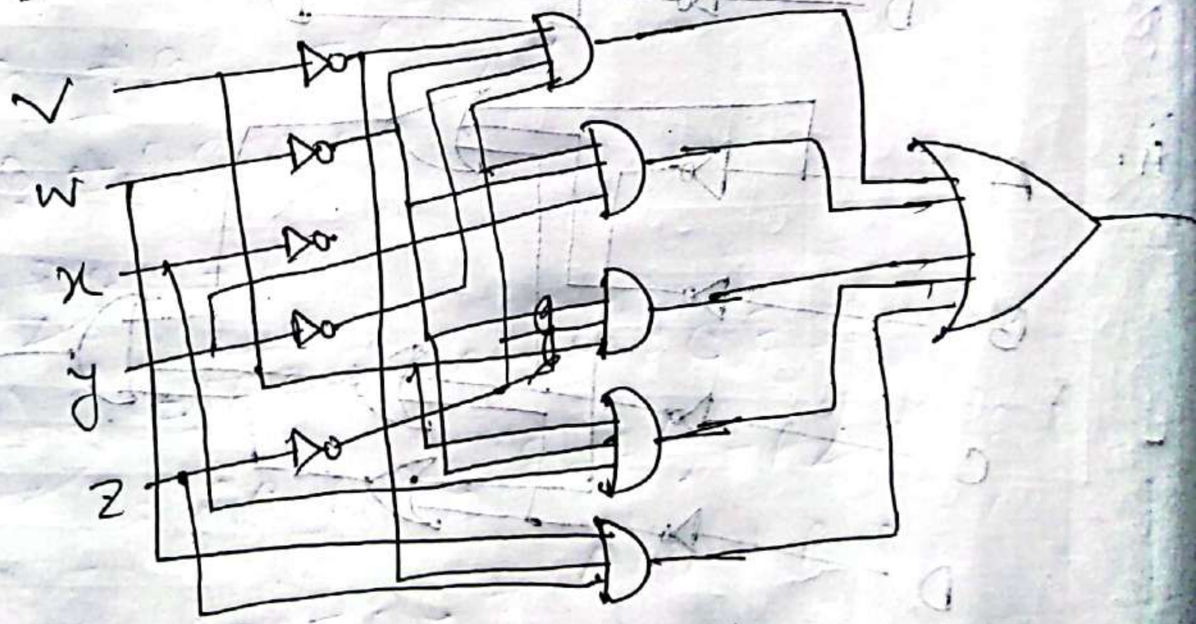
d. i)



ii)



e) i)



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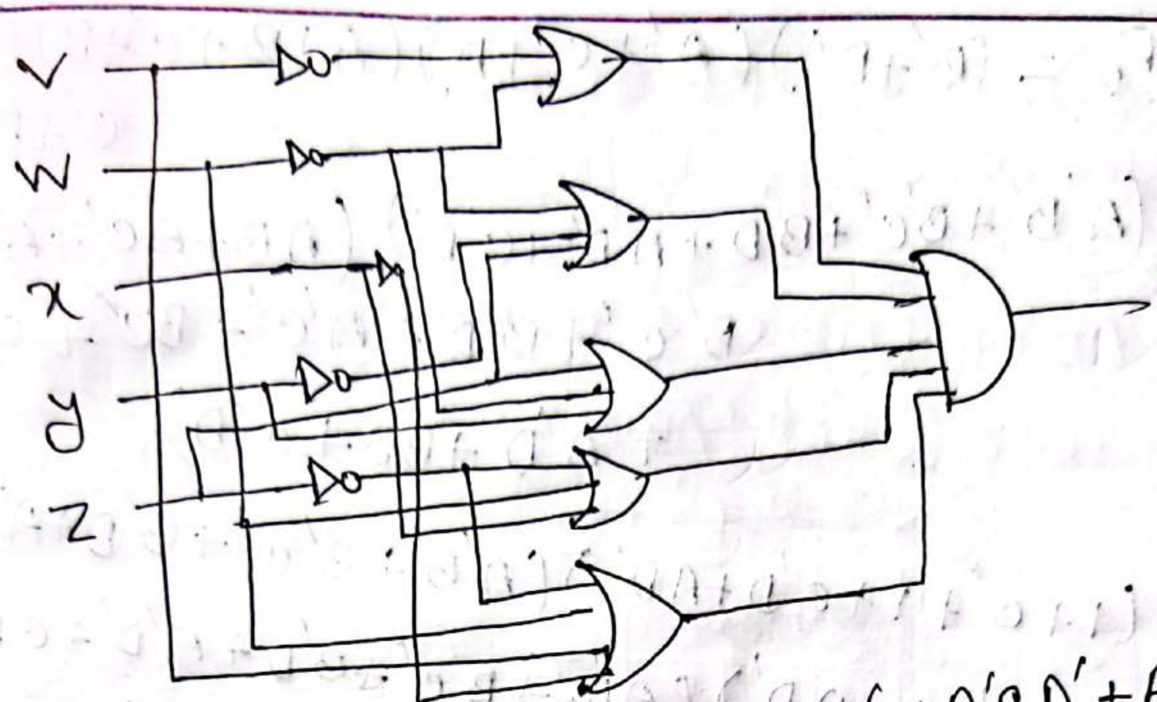


9. a) $F_1 = AC' + ACE + ACE' + A'CD' + A'D'E'$

$= AC(E+E') + AC' + A'(CD' + D'E')$

$= A(C+C') + A'(CD' + D'E')$

$= A + CD' + D'E$

$= ((A + D'(C+E))')'$

$= (A'(D'(C+E))')'$

$= (A'(D + (C+E)')')'$

$= (A'(D + C'E'))'$

$= (A'D + A'C'E')'$

b) $F_2 = (B'+D')(A'+C'+D)(A+B'+C'+D)(A'+B+C'+D')$

$= (A'B'+B'C'+B'D+A'D'+C'D')(AB+AC'+AD'+A'B'C'+B'D'+A'C'+BC'+C'+CD'+A'D+BD+C'D)$

$+A'D+BD+C'D)$ $A'B'+B'C'+B'D+A'D'+C$

$= (1+C'+1+C'D+AD')(A'B'+B'C'+B'D+A'D'+C'D')$

$= (1+C'+1+AD')(A'B'+B'C'+B'D+A'D'+C'D')$

$= (1+1)\ (A'B'+B'C'+B'D+A'D'+C'D')$

$= (A+B')+B'C'+B'D+A'D'$

$= A'B'+B'C'+C'D$

$= ((A'B'+B'C'+C'D')')'$

$= (BD+BC+AC)'$

10. a) $F_1 = AC'+ACE+ACE'+A'CD'+A'D'E'$

$= AC'+AC(E+E')+A'CD'+A'D'E'$

$= AC'+AC+A'CD'+A'D'E'$

$= A(C+C')+A'CD'+A'D'E'$

$= A+A'CD'+A'D'E$

$= A+D'E'+CD'$

b. $F_2 = (B'+D')(A'+C'+D)(A+B'+C'+D)(A'+B+C'+D')$

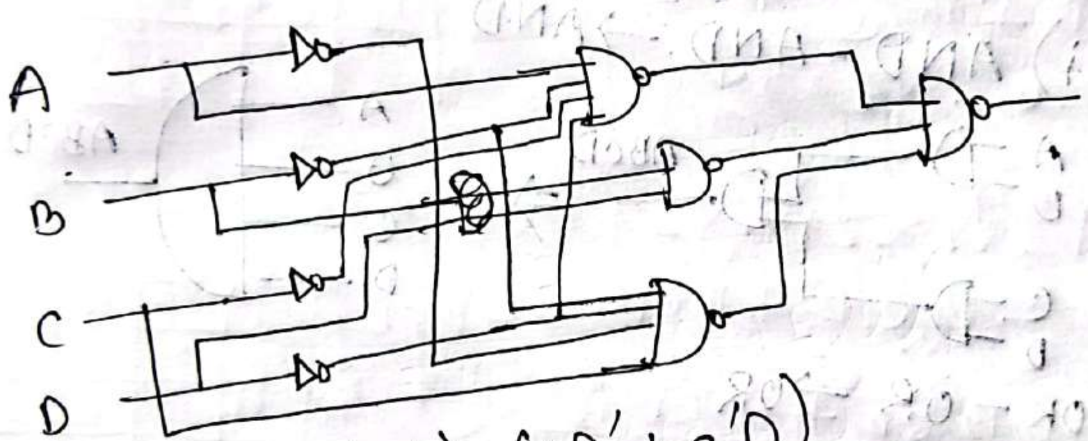
$= A'B' + C'D' + B'C'$

11. a) $BD + BCD + AB'C'D' + A'B'CD'$

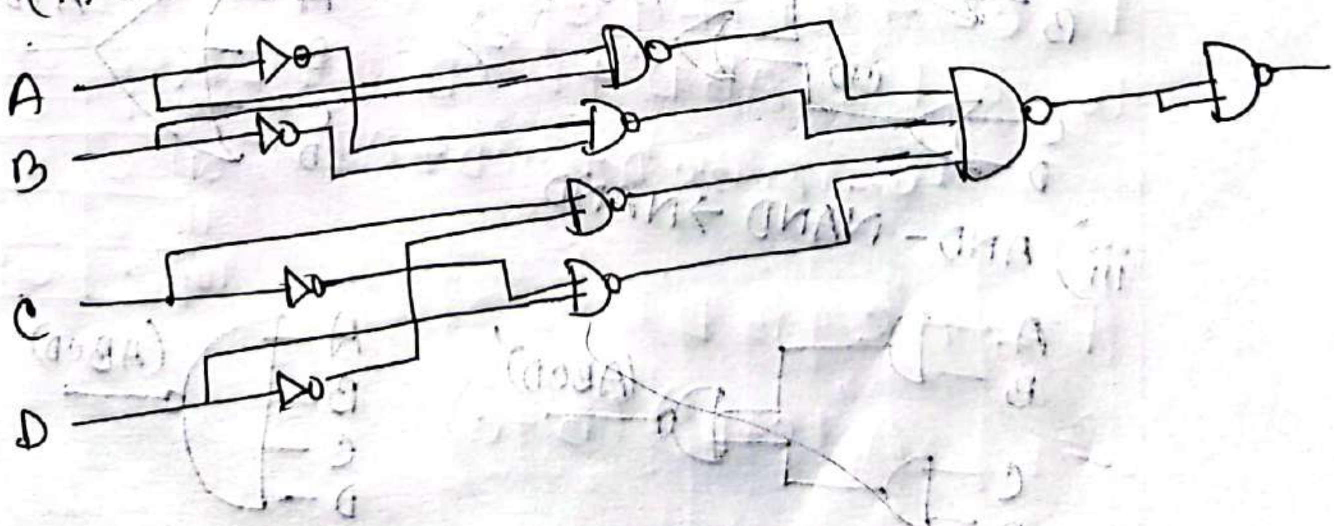
$= BD(1+C) + D'(AB'C' + A'B'C)$

$= BD + D'(AB'C' + A'B'C)$

$= BD + AB'C'D + A'B'CD'$



b) $(AB + A'B')(CD' + C'D)$



$+ DB(AC'$

12. a) $AB' + C'D' + A'CD' + DC'(AB + A'B') + DB(AC' \ldots$

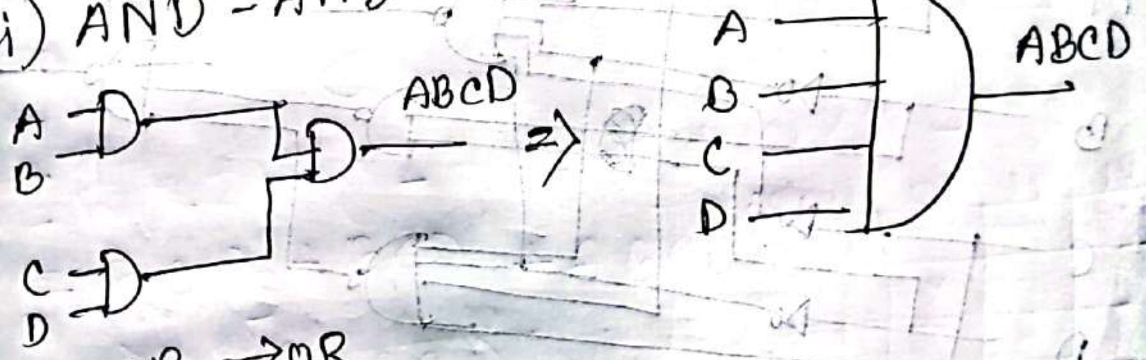
$= (A' + B' + C')(A + B' + C + D')(A + B + C' + D')$

b) $AB'CD' + A'BCD' + AB'C'D + A'BC'D$

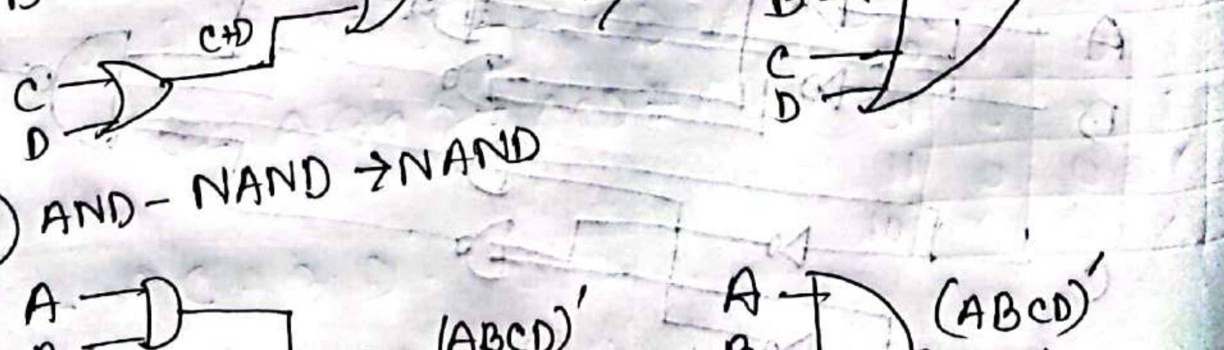
$= (C+D)(C'+D')(A+B)(A'+B')$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	0	0	0
10	0	1	0	1

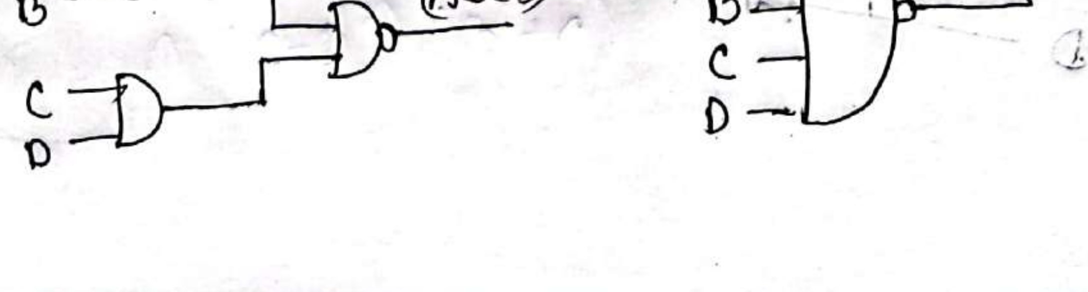
13. i) AND - AND $\rightarrow$ AND



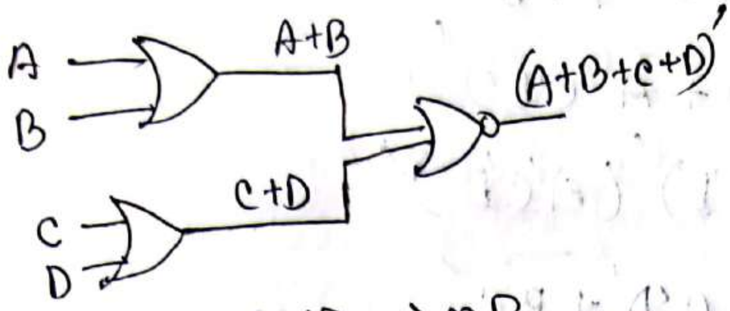
ii) OR - OR $\rightarrow$ OR

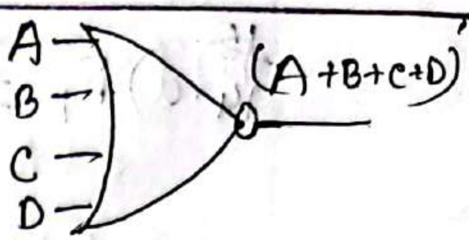


iii) AND - NAND $\rightarrow$ NAND

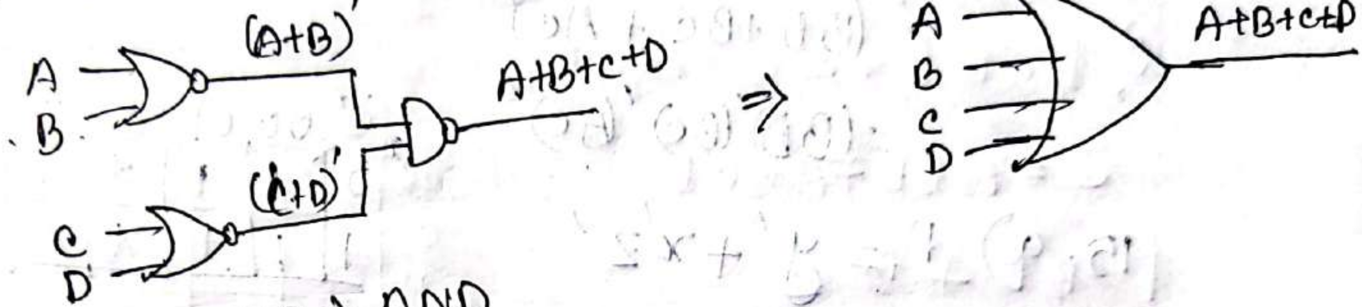


iv) OR — NOR → NOR

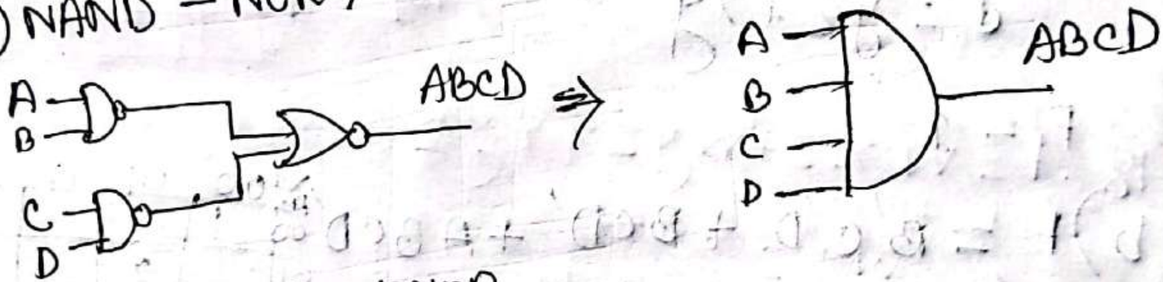




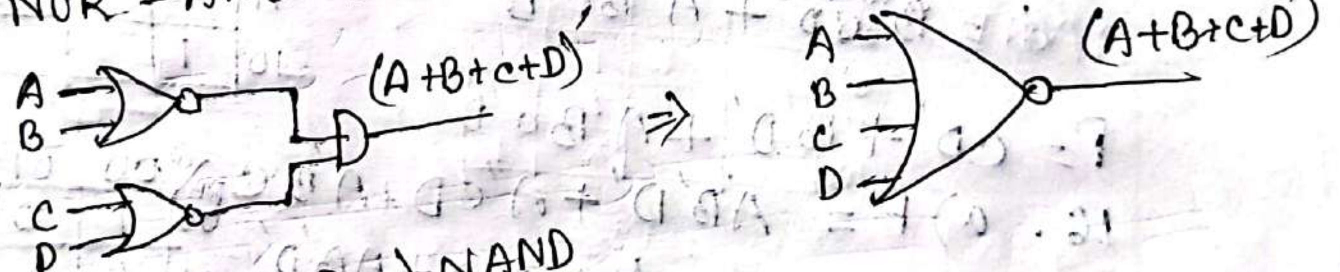
v) NOR — NAND ⇒ OR



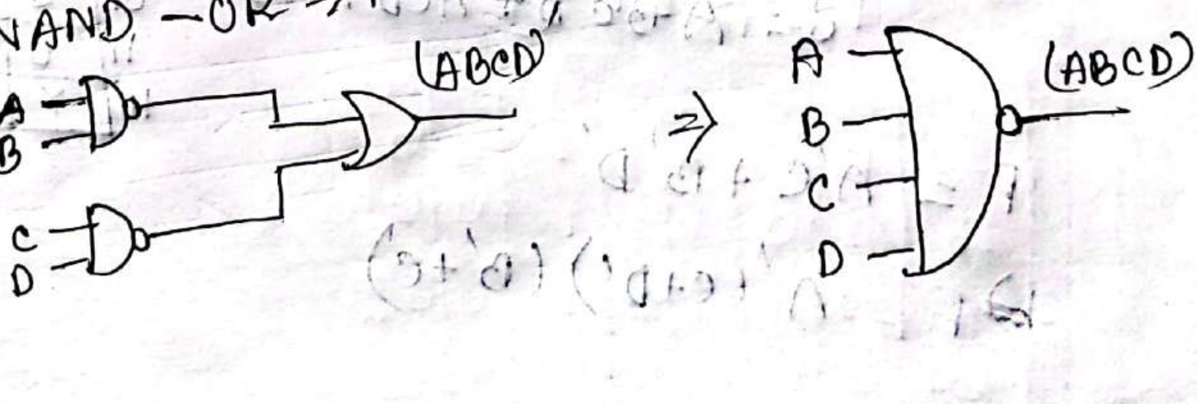
vi) NAND — NOR ⇒ AND



vii) NOR — AND → NOR



viii) NAND — OR ⇒ NAND



15.

14. a) $F_1 = A + D'E' + CD'$

$= (A'D + A'C'E')'$

$= (A'D)'(A'C'E')'$

b) $F_2 = A'B' + C'D' + B'C'$

$= (BD + BC + AC)'$

$= (BD)'(BC)'(AC)'$

$x \backslash yz$	00	01	11	10
0	1	1	x	1
1	1	1	x	x

15. a) $F = y' + x'z'$

$d = yz + xy$

$F = 1$

b) $F = B'C'D' + BCD' + ABCD'$

$d = B'CD' + A'BC'D$

$AB \backslash CD$	00	01	11	10
00	1			x
01		x		1
11				1
10	1			x

$F = CD' + B'D' + A'BC'D$

16. a) $F = A'B'D' + A'CD + A'BC$

$d = A'BC'D + ACD + AB'D'$

$AB \backslash CD$	00	01	11	10
00	1	0	1	1
01	0	X	1	1
11	0	0	X	0
10	X	0	X	X

$F = A'C + B'D'$

b) $F = A'(C + D')(B' + C')$

b) $F = w'(x'y + x'y' + xyz) + x'z'(y+w)$

$= w'x'y + w'x'y' + w'xyz + x'yz' + wx'z'$

$d = w'x(y'z + yz') + wyz$

$= w'xy'z + w'xyz' + wyz$

wx \ yz	00	01	11	10
00	1	1	1	1
01	0	X	1	X
11	0	0	X	0
10	1	0	X	1

i) $F = \cancel{w'z}\, x'z' + w'z$

ii) $F = (x+z)(w'+z')$

c) $F = ACE + A'CD'E' + A'C'DE$

$d = DE' + A'D'E + AD'E'$

AC \ DE	00	01	11	10
00	0	X	1	X
01	1	X	0	X
11	X	1	1	X
10	X	0	0	X

i) $F = AC + CE' + A'C'D$

ii) $F = (C+D)(A'+C)(A+C'+D')$

d) $F = B'D'E' + A'BE + B'C'E' + A'BC'D'$

$d = BDE' + CD'E'$

AB \ CDE	000	001	011	010	110	111	101	100
00	1	0	0	1	1	0	0	X
01	1	1	1	X	X	1	1	X
11	0	0	0	X	X	0	0	X
10	1	0	0	1	1	0	0	X

i) $F = A'B + DE' + B'C'D'E'$

ii) $F = (A'+B')(B+C+E')(B+C'+E')$

$= (A'+B')(B+E')$

17. a) $F = A'B'C' + AB'D + A'B'CD'$

$d = ABC + AB'D'$

AB \ CD	00	01	11	10
00	1	1		1
01				
11			X	X
10	X	1	1	X

$F = AB' + B'C' + B'D'$

$= B'(A + C' + D')$

b) $F = (A+D)(A'+B)(A'+C')$

$= (AB + A'D + BD)(A' + C')$

$= A'D + A'C'D + A'BD + ABC' + BC'D$

$= A'D + ABC'$

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11	1	1		
10				

c) $F = B'D + B'C + ABCD$

$d = A'BD + AB'C'D'$

$F = B'D + B'C$

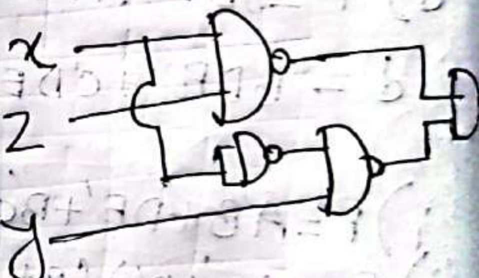
	00	01	11	10
00		1	1	1
01		x	x	
11				
10	x	1	1	1

18. $F = w'xz + w'yz + x'yz' + wxy'z$

$d = wyz$

NAND

wx \ yz	00	01	11	10
00			1	1
01		1	1	
11		1	x	
10			x	1



$F = xz + x'y$

19. $A'BE + BCDE + BC'D'E + A'B'DE' + B'C'DE'$

There are three don't care condition

Here is, $d = ABC'DE + AB'CDE' + ABCD'E$

20. $F = A'B'D' + AB'CD' + A'BD + ABC'D$

AB \ CD	00	01	11	10
00	1			1
01		1	1	
11		1		
10				1

$F = A'BD + BC'D + A'B'D' + B'CD'$

$= BD(A' + C') + B'D'(A' + C)$

21. $f = wxy' + y'z + w'yz' + x'yz'$

$g = (w + x + y' + z')(x' + y' + z)(w' + y + z')$

$= (wx' + wy' + wz + xy' + xz + xy' + y' + y'z + x'z' + y'z')$

$(w' + y + z')$

$= (w'xy' + w'xz + w'xy' + w'y' + w'y'z + w'x'z' + w'y'z' +$

$wxy' + wyz + xyz + x'yz' + wx'z' + wyz' + xy'z' + x'y'z$

$y'z' + x'z' + y'z')$

$= (w'y' + y'z' + x'z' + y'z' + w' + y)$

$= w' + y'z' + x'z' + y)$

$= w' + y'z' + x'z' + y$

$f \cdot g = (wxy' + y'z + w'yz' + x'yz')(w' + y'z' + x'z' + y)$

$= w'y'z + wxy'z' + w'yz' + w'yz' + w'xyz' + w'yz' + w'x'yz' + x'yz' + x'yz')$

$= w'y'z + wxy'z'$

$= w'y'z + w'yz' + wxy'z + w'x'yz' + x'yz'$

$wx \backslash yz$	00	01	11	10
00		1		1
01		1		1
11	1			
10				1

$F = w'y'z + wxy'z' + x'yz' + w'yz'$

22. $xy + x'y'z' + x'yz'$

$= x'z' + xy$

$x \backslash yz$	00	01	11	10
0	1	1	1	
1			1	

$yz \backslash x$	0	1
00	1	
01		
11		1
10	1	1

23. $D(A'+B) + B'(C + AD)$

$= A'D + BD + B'C + AB'D$

$= D + B'C$

$CD \backslash AB$	00	01	11	10
00				
01	1	1	1	1
11	1	1	1	1
10	1			1

$BD \backslash AC$	00	01	11	10
00		1	1	
01	1	1	1	1
11	1	1	1	1
10				

24. a) $F(A, B, C, D, E, F, G) = \Sigma(20, 28, 52, 60$

Table-1

Group	Minterm	A	B	C	D	E	F	G
0	20	0	0	1	0	1	0	0
1	28	0	0	1	1	1	0	0
	52	0	1	1	0	1	0	0
2	60	0	1	1	1	1	0	0

Chapter- 3

24.

Table-2

Group	Pair	A	B	C	D	E	F	G
0	(20,28)	0	0	1	–	1	0	0
	(20,52)	0	–	1	0	1	0	0
1	(28,60)	0	–	1	1	1	0	0
	(52,60)	0	1	1	–	1	0	0

For table-3 repeat same step by taking pairs of succesive groups merging them where there is only a 1-bit difference and keeping them in groups according to the groups from where they are merged and placed - in bit difference.

Table-3

Group	Quad	A	B	C	D	E	F	G
0	20,28,52,60	0	–	1	–	1	0	0

After table 3 process is stopped as there is no 1-bit difference in the remaining group. A'CEF'G' is obtained in table-3 as A, F, G contains 0 so A', F', G', C, and E contains 1 so CE

Prime Implicants Table

Minterm / prime implicants	20	28	52	60
A'CEF'G' (20,28,52,60)	1	1	1	1

Simplified. Boolean. function $= A'CEF'G'$

b) $F(A, B, C, D, E, F, G) = \Sigma(20, 28, 38, 39, 52, 60, 10, 103, 127)$

using tabular method. similar to 2.24

a. $F = ABCDEFG + A'CEF'G' + BC'D'EF$

c) $F(A, B, C, D, E, F) = \Sigma(6, 9, 13, 18, 19, 25, 27, 29, 41, 45, 57, 61)$

$F = A'B'C'DEF' + A'BC'D'E + CE'F + A'BD'EF$

25. a) $F = POS(0, 1, 4, 5)$

Now,

a		b		c	
	xyz		xyz		xyz
0	000	(0,1)	00-	(0,1,4,5)	-0-
1	001	(0,1)	-00	(0,1,4,5)	-0-
4	100	(0,4)	-00	(0,1,4,5)	-0-
5	101	(1,5)	-01		
		(4,5)	10-		

Follow solution 2.24 a) and find $F = y$.

b) F = POS (0, 1, 2, 3, 4, 10, 11)

F(A, B, C, D) = π (0, 1, 2, 3, 4, 10, 11)

a		b		c	
	ABCD		ABCD		ABCD
0	0000	(0,1)	000-	(0,1,2,3)	00--
1	0001	(0,2)	00-0	(0,1,2,3)	00--
2	0010	(0,4)	0-00	(2,3,10,11)	-01-
4	0100	(1,3)	00-1	(2,3,10,11)	-01-
3	0011	(2,3)	001-		
10	1010	(2,10)	-010		
11	1011	(3,11)	-011		
		(10,11)	101-		

NOW,

	0	1	2	3	4	10	11
(A + B) (0,1,2,3)	x	x	x	x			
(B + C') (2,3,10,11)			x	x		x	x
(A + C + D) (0,4)	x				x		

After simplified we find F = (B+C') (A+B) (A+C+D)

c) $F = POS(1, 3, 5, 7, 13, 15)$

a	wxyz	b	wxyz	c	wxyz
1	0001	(1,3)	00-1	(1,3,5,7)	0--1
3	0011	(1,5)	0-01	(1,3,5,7)	0--1
5	0101	(3,7)	0-11	(5,7,13,15)	-1-1
7	0111	(5,7)	01-1	(5,7,13,15)	-1-1
13	1101	(5,13)	101-		
15	1111	(7,15)	-111		
		(13,15)	11-1		

NOW,

		1	3	5	7	13	15
$(w+z')$	(1,3,5,7)	x	x	x	x		
$(x'+z')$	(5,7,13,15)			x	x	x	x

$\therefore f = (w + z')(x' + z')$

26. a. $F = ACE + A'CD'E + A'C'DE$

$d = DE' + A'D'E + AD'E$

b. $F = B'DE' + A'BE + B'C'E' + A'BC'D'$

$d = BDE' + CD'E'$

a) $F = ACE + A'CD'E' + AC'D'E' + A'C'DE$

$= \Sigma(3, 4, 13, 15)$

$d = DE' + A'D'E + AD'E'$

$= ACDE + AC'DE' + A'CDE' + A'C'DE' + A'C'D'E + ACD'E' + AC'D'E'$

$= \Sigma(14, 10, 6, 2, 5, 1, 12, 8)$

$= \Sigma(1, 2, 5, 6, 8, 10, 12, 14)$

NOW

a

	ACDE
1	0001
2	0010
4	0100
8	1000
3	0011
5	0101
6	0110
10	1010
12	1100
813	1101
14	1110
15	1111

b

	ACDE
(1,3)	00-1
(1,5)	0-01
(2,3)	001-
(2,6)	0-10
(2,10)	-010
(4,5)	010-
(6,6)	01-0
(4,12)	-100
(8,10)	10-0
(8,12)	1-00
(5,13)	-101
(6,14)	-110
(10,14)	1-10
(12,12)	11-0
13,15	11-1
14,15	111-

c

	ACDE
(2,6,10,14)	--10
(2,6,10,14)	--10
(4,5,12,13)	-10-
(4,5,12,13)	-10-
(4,6,12,14)	-1-0
(4,6,12,14)	-1-0
(8,10,12,14)	1--0
(8,10,12,14)	1--0
(12,13,14,15)	11--
(12,13,14,15)	11--

Now,	3	4	13	15
$A'C'E$ (1,3)	x			
$A'B'E$ (1,5)	x			
$A'C'D$ (2,3)				
DE (2,6,10,14)		x	x	
CD' (4,6,12,14)		x		
CE' (4,6,12,14)				
AE' (8,10,12,14)			x	x
AC (12,13,14,15)				

$F = A'C'E + CD' + AC$

d) $F = B'DE' + A'BE + B'C'E' + A'BC'D'$

$+ A'B'C'DE' + A'BCDE +$

$= AB'CDE' + AB'C'DE + A'B'CDE' + A'B'C'D'E' + A'BC'DE + A'BC'D'E$

$A'BCD'E + A'BC'DE + A'B'C'D'E' + A'BC'DE + A'BC'D'E$

$= \Sigma(0, 2, 6, 8, 9, 11, 13, 15, 16, 18, 22)$

$d = ABCDE' + ABC'DE' + A'BCDE' + A'BC'DE' +$

$ABCD'E' + AB'CD'E' + A'BCD'E' + A'B'CDE'$

$= \Sigma(4, 10, 12, 14, 20, 26, 28, 30)$

using tabular meth, $F = B'E' + A'B$.